

Aditya Nandan

nandanaditya985@gmail.com · +91 8439336494 · github.com/adityanandanx · linkedin.com/in/aditya-nandan-thats-it

Education

Graphic Era Deemed to be University, Dehradun

B.Tech in Computer Science Engineering

June 2027

GPA 8.54/10

Internship Experience

Quivr (YCombinator-backed) – Frontend Intern & Open Source Contributor

May 2023 – November 2024

Remote

- Built the notetaker feature from scratch end-to-end (8 PRs, 6 merged) using TipTap/ProseMirror, shipping a formatting toolbar, sticky search bar, and scoped styling as Notetaker 0.1 (#3097, #3111, #3117, #3258)
- Users needed to query their knowledge base without leaving the editor; implemented "@"-mention brain selection and a Mod+F search bar, and built a shared ChatBar component reused across search and notetaker, extending the chat/session model to support note-scoped conversations (#3215)
- AI-generated text was inserted directly into notes with no way to reject it; designed a custom TipTap AIHighlight extension marking AI output as provisional, with explicit accept/decline actions before it entered the document (#3389)
- Knowledge bases were single and implicit, blocking per-use-case organization; designed a multi-brain architecture and shipped a searchable brain switcher on Radix UI popovers with inline creation and deletion, after iterating past an unmerged localStorage-based prototype (#309 prototype, #452, #456, #507)
- Restructured the initial prototype to the Next.js App Router, built the first reusable primitives (Button, NavBar, Modal, Toast, Tooltip), and shipped multi-conversation chat with Supabase-backed history via a React context layer after the initial hook caused duplicate API calls (#55, #123, #163, #252, #290, #297)

Projects

Heimdall: Local AI Screen Memory for Hyprrland (Wayland)

github.com/adityanandanx/heimdall/

- Event-driven Python capture daemon on the Hyprrland socket: on window changes, captures the active-window region (grim) and its accessibility tree (AT-SPI) into a searchable SQLite/FTS5 store; falls back to RapidOCR on the Intel NPU (OpenVINO) with CPU fallback for blind windows, and indexes YouTube watch-sessions (MPRIS + yt-dlp/faster-whisper transcripts) via subtitles
- "Day recap" and "time breakdown" pipes (APScheduler) run through a local Gemma 4 QAT model on the Arc Vulkan backend (llama-server); FastAPI backend with a Tauri 2 desktop client for day timeline, search, sessions, status and settings – everything stays on-machine

Hybrid RAG pipeline: Conversational Movie Agent

github.com/adityanandanx/cinema-rag

- Built a multi-agent hybrid Retrieval Augmented Generation (RAG) pipeline answering natural language queries across 2200 movie screenplays chunked into 289124 documents, combining dense retrieval (BAAI/bge-small-en-v1.5) with sparse BM25 retrieval via Reciprocal Ranked Fusion (RRF), Qdrant as the vector store, and Llama 4 17b via Groq for fast response generation

Nirbhay: Automated Emergency Detection App

github.com/Team-Hacksmith/nirbhay

- Built an accessibility-focused safety app that automatically detects distress and triggers hands-free emergency alerts, integrating wearable hardware (ESP32), a Flutter mobile app, and a Firebase backend with YAMNet for real-time audio detection – empowering elderly, disabled, and at-risk populations to summon help without manual intervention

Skills

- **Languages:** Python, TypeScript, JavaScript, Java, C, C++, PHP
- **AI/ML:** LangChain, LangGraph, Hybrid RAG, Agentic AI, Generative AI, Computer Vision, NLP
- **Frontend:** React.js, Next.js, Vue.js, Flutter, Tailwind CSS, SCSS
- **Backend:** FastAPI, Flask, Django, Laravel, Node.js, Express.js
- **Databases & Cloud:** PostgreSQL, MySQL, MongoDB, Firebase, AWS (S3, EC2, Lambda), Git, Docker

Honors and Awards

WorldSkills Asia: Web Technologies

Nov 2025

WorldSkills Asia Taipei, Taiwan

- Represented India in Web Technologies against 14 countries across 5 independent 3-hour modules (speed test, frontend, backend, fullstack, team app) over 3 days, each with a fresh spec and zero internet access; only offline documentation provided
- Built each module from scratch to a pixel-accurate design and marking-scheme test cases within the 3-hour window using Laravel, Next.js, React, Vue, and Nuxt, relying solely on offline docs with no external references or libraries
- Won Medallion of Excellence, ranked 5th globally

IndiaSkills National Qualifier: Web Technologies

2024

Central Government of India, Ministry of Skill Development & Entrepreneurship

- Underwent 2+ months of intensive training and 3 internal test rounds simulating competition constraints (fixed time, blind specs, no external resources)
- Won IndiaSkills National 2024 in Web Technologies, securing selection to represent India internationally